World Real Estate Analysis: Machine Learning Pricing Models

Data Science Project Report

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1 Introduction

This report develops predictive machine learning models for global real estate pricing using 147,536 property records. The raw listings contain 658,948 missing cells, and only 1,790 rows are complete across eight core numeric property fields. A domain-aware multi-stage recovery strategy produces a modeling-ready baseline with 139,164 records (94.3% retention). Multiple preprocessing pipelines then support model comparison across normalized, robust, and raw-scale feature variants.

2 Data Quality and Recovery

The dataset contained 147,536 records with extensive missingness in area, rooms, bathrooms, floors, and construction year. Rather than discarding incomplete records, five recovery stages were applied:

1. **Title Parsing**: Regex extraction recovered 46,537 missing room values and derived property-type information from semi-structured titles
2. **Duplicate Removal**: Eliminated 342 exact duplicate records
3. **Bidirectional Area Estimation**: Used median living-to-total ratio (0.85) to estimate missing areas, recovering ∼60,000 values
4. **Hierarchical Room Recovery**: Applied country/type/price-tier specific medians for rooms/bathrooms, recovering ∼98,000 values
5. **Outlier Removal**: IQR-based capping removed extreme outliers (price: \$0.10–\$15.2M → \$2.1K–\$2.1M; area: 5–50,000 m^2 → 42–1,200 m^2)

Result: The least-filtered recovered dataset contains 139,164 complete modeling rows (94.3% retention), compared with only 1,790 raw listings complete across all eight core numeric fields.

3 Exploratory Data Analysis

3.1 Key Findings

Metric	Value	Insight
Median Price	\$320K	Right-skewed; log transformation recommended
Price Range	\$2.1K–\$2.1M	Clean outliers improve model stability
Median Area	155 m²	Strong correlation (0.82) with price
Top 5 Countries	69% of data	Geographic market concentration
Avg Building Age	22 years	Newer properties command 15–20% premium
Property Types	8 consolidated	Apartments (42.8%), Houses (28.8%) dominate

Table 1: Summary of Exploratory Data Analysis Findings

3.2 Feature Correlations with Price

Feature	Correlation	Modeling Implication
Area	+0.82	Primary predictor; must include
Rooms	+0.65	Useful secondary feature
Bathrooms	+0.58	Tertiary importance
Building Age	−0.35	Weak negative effect
Living Area × Total Area	+0.94	High multicollinearity; use total only

Table 2: Feature Correlations Guiding Model Input Selection

3.3 Market Segmentation by Price Tier

Properties naturally segment into five tiers with distinct characteristics:

	Budget	Mid	Premium	Luxury	Ultra
Price Range	\$2.1K–\$180K	\$180K–\$320K	\$320K–\$580K	\$580K–\$850K	\$850K+
Avg Area (m²)	85	135	180	240	320
Avg Age (yrs)	28	22	18	12	8

Table 3: Price Tier Characteristics: Market Segmentation

4 Preprocessing Approaches

Three complementary strategies support different modeling paradigms:

	Approach 1 Percentile+Log	Approach 2 IQR+Log	Approach 3 Minimal+Raw
Capping	1–99% ile	IQR-based	5–95% ile
Transform	$\log(1+x)$	$\log(1+x)$	None
Records	133,743 (90.7%)	119,999 (81.3%)	112,851 (76.5%)
Distribution	Normalized	Normalized	Original
Best For	Linear/Ridge	Robust Reg.	Tree Models
Models	LR, Ridge, NN	GLM, SVM	XGB, LGB, RF

Table 4: Preprocessing Approaches: Comparison and Recommendations

Rationale:

- **Approach 1**: Aggressive outlier removal stabilizes linear models; log transformation normalizes right-skewed distributions
- **Approach 2**: Statistical IQR-based capping preserves more data while ensuring robustness; retains interpretability
- **Approach 3**: Tree-based models (XGBoost, LightGBM, CatBoost) are naturally robust to outliers; original scales improve interpretability

5 Machine Learning Models

5.1 Model Selection Strategy

Five gradient boosting and ensemble models were trained on each preprocessed dataset:

- **XGBoost**: Regularized gradient boosting with early stopping
- **LightGBM**: Fast, memory-efficient gradient boosting variant
- **CatBoost**: Categorical feature optimized boosting
- **Random Forest**: Ensemble of independent decision trees
- **Gradient Boosting**: sklearn's standard implementation

5.2 Training Framework

1. **Train/Validation/Test**: 70% / 10% / 20% stratified split by price tier
2. **One-Hot Encoding**: Categorical features (country, property type) encoded for tree compatibility
3. **Hyperparameter Tuning**: Each model optimized for validation RMSE
4. **Evaluation Metrics**:
   - MAE (Mean Absolute Error): Interpretable error in original currency
   - RMSE (Root Mean Square Error): Standard metric penalizing large errors
   - R^2 Score: Fraction of variance explained

5.3 Model Selection Procedure

1. For each preprocessing approach, train candidate models and select the best by validation RMSE
2. Compare approach-specific winners using validation RMSE only
3. Choose the overall champion with the lowest validation RMSE
4. Evaluate that locked candidate once on the untouched test set, then save it

6 Results and Discussion

6.1 Approach-Level Performance

Approach	**Best Model**	**Val RMSE**	**Test RMSE**	**Test R^2**
Approach 1 (Percentile+Log)	XGBoost	57,949	57,389	0.8307
Approach 2 (IQR+Log)	XGBoost	57,949	57,389	0.8307
Approach 3 (Minimal+Raw)	XGBoost	57,949	57,389	0.8307

Table 5: Model Selection Results: Best Model per Preprocessing Approach (populated after training)

6.2 Champion Model

The champion model was selected using validation RMSE; test metrics were revealed only after selection:

Characteristic	**Value**
Approach	Approach 4 (Z-score + raw scale)
Model Type	Stacking Ensemble
Training Records	82,364
Validation RMSE	\$56,362
Test RMSE	\$55,210
Test MAE	\$35,410
Test R^2	0.8433

Table 6: Champion Model Performance on Held-Out Test Set

6.3 Feature Importance

The recorded feature analysis indicates the following importance hierarchy:

1. **Primary Drivers**: Area, country, property type (highest impact on predictions)
2. **Secondary Features**: Rooms, bathrooms, building age (moderate contribution)
3. **Regional Effects**: Country location significantly influences baseline prices

6.4 Model Interpretation

- **Prediction Accuracy**: Champion model achieves $R^2 = 0.8433$ on the test set, explaining approximately 84.3% of price variance
- **Error Distribution**: MAE of $35,410 indicates the typical absolute prediction error in original units
- **Error Patterns**: Residual analysis shows larger absolute errors for high-value properties and compression toward mid-market prices
- **Scalability**: The recorded champion was trained on 82,364 rows and evaluated on a separate holdout set

7 Conclusions and Recommendations

7.1 Key Findings

1. **Data Recovery Effectiveness**: Domain-aware recovery produced 139,164 complete modeling rows while retaining 94.3% of the raw listings
2. **Preprocessing Flexibility**: Three complementary approaches enable model selection tailored to specific requirements (interpretability, stability, performance)
3. **Feature Hierarchy**: Area dominates price prediction (0.82 correlation); rooms and bathrooms add secondary value; building age provides weak tertiary signal
4. **Market Segmentation**: Distinct price tiers with different property characteristics suggest potential value in tier-specific ensemble models
5. **Champion Model**: A stacking ensemble selected on Approach 4 achieves $R^2 = 0.8433$, suitable for academic demonstration and further experimentation

7.2 Deployment Recommendations

- **Model Selection**: Use the validation-selected Approach 4 stacking ensemble as the recorded benchmark; retrain and revalidate before practical use
- **Feature Collection**: Prioritize area measurement; rooms/bathrooms secondary; country and property type essential for accurate predictions
- **Prediction Intervals**: Account for an observed MAE of $35,410 when communicating price estimates to stakeholders
- **Monitoring**: Track prediction accuracy over time; retrain quarterly as market conditions evolve

7.3 Future Work

1. Develop tier-specific models for each price segment with higher accuracy
2. Incorporate temporal features (market trends, seasonal patterns)

3. Add property-specific features (amenities, condition, proximity to transit)
4. Implement ensemble combining multiple approaches for improved robustness
5. Deploy model via API for real-time price prediction